ALAN GUTIERREZ RAMIREZ

C++ Software Developer · Graphics & Simulation

Valencia, Spain · +34 603 586 727 · alangutier24@gmail.com · LinkedIn · Portfolio

PROFESSIONAL SUMMARY

C++ Software Developer with 3+ years of experience in graphics programming, real-time simulation, and engine architecture. Proven track record delivering production-grade tools and physics-based simulation systems for industrial training applications. Hands-on expertise across the full graphics pipeline using OpenGL and PhysX, with strong foundations in linear algebra, memory management, and multithreading. EU citizen based in Valencia — available for on-site, hybrid, or fully remote roles across Europe.

TECHNICAL SKILLS

Languages: C++, C, C#, Python, GLSL/HLSL, SQL

Graphics & Physics: OpenGL, PhysX, DirectX, Real-Time Rendering, 3D Math / Linear Algebra, Vulkan (familiar)

Engines: Custom Game Engines, Unreal Engine 5, Unity

Tools & Version Control: Visual Studio, Git, Perforce, CMake, RenderDoc

Core Competencies: OOP, Memory Management, Multithreading, Engine Architecture, Debugging & Optimization, QA

WORK EXPERIENCE

Junior Software Developer | **LSyM – Laboratorio de Simulación y Modelado** Valencia, Spain · Nov 2024 – Oct 2025

C++ · PhysX · SQL · OpenGL · Real-Time Simulation

- Engineered a C++ test-creation tool for heavy machinery training simulators, directly improving instructor workflow efficiency and user assessment capabilities.
- Contributed to a real-time physics-based simulation platform, implementing and optimizing core engine mechanics using PhysX to ensure accuracy and performance.
- Designed and implemented a scalable SQL database schema for simulation data management and retrieval across training scenarios.
- Performed rigorous QA, debugging, and code optimization on engine mechanics and physics models, accelerating project delivery timelines.
- Translated and standardized complex technical manuals (Spanish to English), enabling international deployment of simulation products.

KEY PROJECTS

Graphics Engine Developer | **JAM Engine — Custom C++ / OpenGL Engine** Valencia, Spain · Portfolio Project

- Architected a custom 3D graphics engine from scratch in C++ with OpenGL, including full render loop, memory management, and ECS architecture.
- Implemented advanced rendering techniques: real-time lighting, shadow mapping, volumetric lighting, and particle systems.
- Integrated sound manager and produced comprehensive technical documentation covering API usage and code structure.

Game Programmer | **Shelley Manor — PC Game (Steam)** Valencia, Spain · Student Game Studio

🏆 **Best Student Game · Latam Games Com 2025** | 🏅 **Top 5 Best Hobby Game · GDWC Winter Season**

- Developed and shipped a 3D narrative game on Steam using Unreal Engine 5 and C++, collaborating within a multidisciplinary team.
- Implemented gameplay systems, camera controls, lighting, materials, and post-processing effects.

- Managed debugging and rendering optimization to meet production milestones.

EDUCATION

HND in Video Game Programming | **ESAT – Escuela Superior de Arte y Tecnología**Valencia, Spain · Completed July 2024

Focus: Advanced C++, Graphics Programming, Game Engine Architecture.

Computer Science Coursework · Gustavus Adolphus College, USA2014 – 2017

LANGUAGES & WORK AUTHORIZATION

Spanish: Native · **English:** Fluent (Professional Working Proficiency) · **EU Citizen (Spain) — No sponsorship required**